ISOM SHEAVES IN THE QUOTIENT STACK IN TERMS OF THE GROUPOID

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 044R: sources/stacks-chow/tags.txt, source lines 5410–5410. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 044P: sources/stacks-chow/tags.txt, source lines 5408–5408. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Assume $B \to S$, (U, R, s, t, c) and $\pi : \mathcal{S}_U \to [U/R]$ are as in Lemma 044R. Let S' be a scheme over S. Let $x, y \in \mathrm{Ob}([U/R]_{S'})$ be objects of the quotient stack over S'. If $x = \pi(x')$ and $y = \pi(y')$ for some morphisms $x', y' : S' \to U$, then*

$$Isom(x, y) = S' \times_{(y', x'), U \times_S U} R$$

as sheaves over S'.

Proof. Let $[U/_pR]$ be the category fibred in groupoids associated to the presheaf in groupoids (044P) as in the proof of Lemma 044R. By construction the sheaf $Isom(x, y)$ is the sheaf associated to the presheaf $Isom(x', y')$. On the other hand, by definition of morphisms in $[U/_pR]$ we have

$$Isom(x', y') = S' \times_{(y', x'), U \times_S U} R$$

and the right hand side is an algebraic space, therefore a sheaf. □

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